# Lesson 22: Modelling with Sinusoidal Functions — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

## Section 1 — Warm-up (foundation)

**1.** Max of y = sin x?

**2.** Min of y = sin x?

**3.** Period of y = sin x (degrees)?

**4.** Amplitude when max = 9, min = 1?

**5.** Midline when max = 9, min = 1?

**6.** Amplitude when max = 5, min = −5?

## Section 2 — Core practice

**7.** Model has max 17, min 3. Find a and d.

**8.** A graph repeats every 180°. Find b.

**9.** Find b if the period is 90°.

**10.** a = 6, d = 10. Find the max and min.

**11.** Max 12, min 4. Find a and d.

**12.** Find b if the period is $2\pi/3$ (radians).

## Section 3 — Challenge & exam-style

**13.** A tide model $y = a \sin(b(t-c)) + d$ has max 8 m, min 2 m, period 12 h. Find a, d and b (degrees).

**14.** A wheel model has max height 20 m, min 2 m, period 40 s. Find a, d and b (degrees).

**15.** Temperature varies between 6°C and 18°C over 24 h. Find a, d and b (degrees).

**16.** A model has amplitude 4, midline 10, period 90°. Write $y = a \sin(bx) + d$.